

Crunch: Standard Tool Guide

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Crunch

Crunch is the program that constructs ad-hoc wordlists that are usually used in password cracking or in penetration testing exercises. It allows users to tweak certain options such as the minimum and maximum length of passwords and the type of characters to appear in the wordlist, and thereafter builds it. The output of the tool is a deduced word list (generated), that meets the given pattern or requirement provided by the user.

Setup and Launch

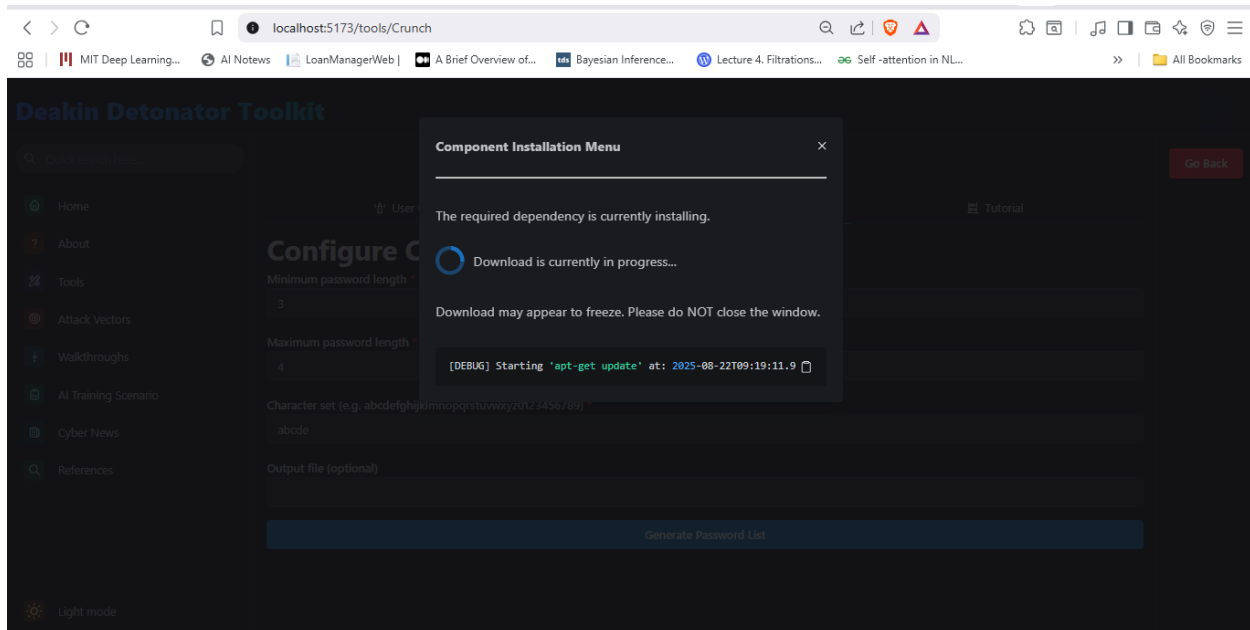
The Deakin Detonator Toolkit (DDT) provides a user interface in which a user can perform relevant preparations and the execution of Crunch. The user can start the DDT system from code using one of the following command lines as per his/her development environment:

yarn run tauri dev

npm run tauri dev

This command builds and runs that the GUI form of the DDT tool in a local, safe execution context byTauri. It also provides the user with consoles interface notifications in real time, where the user is provided with system level messages. The Crunch tool in DDT checks and installs dependencies before it can be used, as illustrated in the screenshot below

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Step-by-step guide

The Crunch user interface is simple and provides the necessary fields to configure the parameters for the generated wordlist. The principal fields are:

1. Minimum password length: This field requires a number and defines the minimum length for the passwords that will be generated.
2. Maximum password length: This field requires a number and defines the maximum length for the passwords that will be generated.
3. Character set (e.g., abcdefghijklmnopqrstuvwxyz0123456789): In this field, you specify the characters you want to use to build the passwords. This can include letters, numbers, and symbols.
4. Output file (optional): You can enter a filename here to automatically save the generated wordlist.

Once you fill the fields, you could press the Generate Password List button to execute the process by clicking the “Generate Password List” button.

Configurations

The system setup of Crunch can be done by way of the UI.

Flags and functions

The most important user inputs are the following:

Field	Description	Example
Minimum password length	Required. A number that sets the shortest possible length for the generated passwords.	3
Maximum password length	Required. A number that sets the longest possible length for the generated passwords.	4
Character set	Required. The string of characters to be used to create the wordlist.	abcde
Output file	Optional. The name of the file to which the wordlist will be saved.	passwords.txt

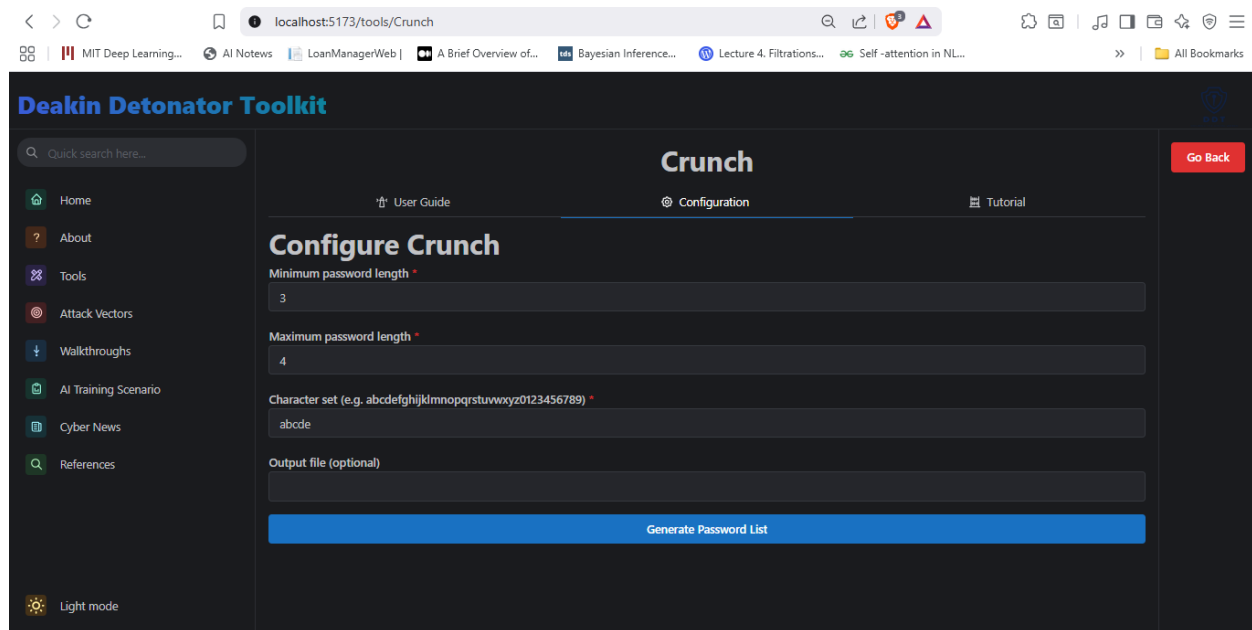
Expected output fields

The expected output is a list of all possible combinations of passwords with respect to the parameters specified by the user, which would be highlighted in the section below the main configuration panel. The list would also be saved to a specified output file unless no such file was specified, in which case it would not be saved.

Tool Usage

UI walkthrough

The DDT UI for the Crunch password generation tool contains two panels: the configuration panel with the fields that the user can set to specify the constraints for the wordlist generated, and the output panel for the generated list, as shown in the screenshot below.



Example

To illustrate this, say we want any number of characters long between 3 and 4 characters in the character set of 'abc'. In this case Crunch would produce all the possible combinations of the given character length--in this case all the 3-character and 4-character combinations of a, b, and c.

Output and Interpretation

The result would be a list that consists of:

- aaa
- aab
- aac
- ...

This list would go on, up to ccc, and then continuing with all combinations of length 4, such as aaaa and aaab until all the constraints are met.